# 1 This Week's Progress

## 1.1 General Items

- Completed Phase 2 Latin Hypercube Sampling (LHS) data generation for the $PtcO_2$-to-$PaO_2$ physics-informed machine learning project.
- Placed a purchase order for the FTDI NMC-2.5M USB null modem cable required for serial communication with the Masimo Rad-97 Pulse CO-Oximeter.
- Awaiting a code summary from a collaborator covering PPG-to-skin-color analysis, to be integrated into the broader skin tone confound investigation.
- Documented current hardware and simulation progress for internal reference and upcoming advisor review.

## 1.2 Technical Progress

- **Phase 2 COMSOL/ML Sweep:** Successfully completed the full Phase 2 simulation batch – 3,000 runs (500 LHS samples × 6 $PaO_2$ levels: 50, 65, 80, 95, 110, and 130 mmHg) – with occlusion disabled (`d_rate = 0`) and the solver running to 800 s. Results are saved with checkpoint-style crash-safe logic (CSV every sample, full `.mat` every 25 samples).
- **Rad-97 Hardware:** Identified that the Rad-97's USB port is a host-type USB-A port, requiring a null modem bridge cable rather than a standard USB cable. Ordered the correct FTDI NMC-2.5M cable (dual FT232R, VID 0403 / PID 6001) to enable serial data capture via PuTTY or Tera Term using the IAP protocol.
- **Skin Tone Estimation from PPG (Direction 1):** Completed review and integration of the PPG-to-skin-tone analysis pipeline (`direction_1_ppg_to_skin_tone/`). Full methodology and results are described in the subsection below (Section 3).

## 1.3 Challenges / Notes

- **Current issue:** The Rad-97 live data capture pipeline cannot be validated until the FTDI NMC-2.5M cable arrives and VCP driver installation is confirmed.
- **Impact:** Hardware testing and real-time $PtcO_2$/$SpO_2$ data acquisition are on hold pending cable delivery.
- **Plan:** Pursue forward COMSOL model design exploration and ML architecture scoping in parallel; finalize Rad-97 setup immediately upon cable arrival.

## 2 Weekly Plans

- Set up the Masimo Rad-97 for live serial data capture once the FTDI cable arrives; verify IAP protocol communication and log a test session via PuTTY or Tera Term.
- Explore design options for the forward COMSOL model – define the target input/output structure and candidate modeling approaches for $PaO_2$ prediction.
- Survey ML model architectures (e.g., physics-informed neural networks, surrogate/emulator models) that could be matched to forward COMSOL simulation outputs; assess feasibility of coupling with the existing Phase 2 inverse pipeline.
- Investigate skin tone data sources and experimental design options to expand the $r_{DC}$-based melanin proxy validation beyond the current dataset.
- Prepare updated documentation and plots from Phase 2 simulation results for advisor review.

# 3 Direction 1: Skin Tone Estimation from PPG Signal

## 3.1 Background and Motivation

Pulse oximeters illuminate tissue with two wavelengths of light:

- **660 nm (red)** — strongly absorbed by deoxygenated hemoglobin (deoxyHb)
- **905 nm (near-infrared)** — comparatively insensitive to hemoglobin oxygenation state

Before reaching the hemoglobin in blood, both wavelengths must pass through the skin. Melanin — the pigment responsible for skin color — absorbs more light at 660 nm than at 905 nm. Therefore, the ratio of the two DC (baseline, non-pulsatile) signals should carry information about the amount of melanin present. We define:

$$r_{DC} = \frac{DC_{\text{red}}}{DC_{\text{IR}}} \tag{1}$$

**Hypothesis:** Darker skin → more melanin → greater red light absorption → lower $r_{DC}$.

## 3.2 The Confound: Hemoglobin Masks the Melanin Signal

The 660 nm channel is also where deoxyHb absorbs strongly — approximately 10× more than oxygenated hemoglobin (oxyHb). When $SaO_2$ varies, rising deoxyHb absorbs more red light and drives $r_{DC}$ downward, mimicking the optical signature of darker skin. Naively using all data allows $SaO_2$ variation to dominate and overwhelm the melanin signal.

**Solution — Normoxic Filter:** Analysis is restricted to time windows where $SaO_2 \geq 94\%$. With hemoglobin locked in its oxygenated state, deoxyHb absorption is minimized, and residual variation in $r_{DC}$ is driven primarily by melanin absorption rather than oxygenation state.

| **Filter Applied** | **4-ch Spearman $\rho$ ($r_{DC}$ vs. Monk scale)** |
|---|---|
| No filter (all SQI-passing windows) | $-0.510$ |
| $SaO_2 \geq 94\%$ (normoxic only) | $-0.623$ |

Table 1: Effect of normoxic filter on the correlation between $r_{DC}$ and Monk skin tone.

## 3.3 Interpreting the Spearman Correlation

Spearman $\rho$ measures how consistently two variables move in the same direction using rank order rather than raw values, making it appropriate here because the relationship between melanin concentration and light absorption follows Beer–Lambert physics (approximately logarithmic, not linear). It ranges from $-1$ (perfectly inverse) to $+1$ (perfectly direct).

A value of $\rho = -0.623$ means: encounters with darker skin reliably tend to have lower $r_{DC}$, and this relationship is moderately strong. The negative sign is expected by the melanin absorption hypothesis. A perfect $-1.0$ would require every darker patient to have a strictly lower $r_{DC}$ than every lighter patient, which is physically unrealistic given noise, finger geometry variation, and probe placement. Achieving $\rho = -0.623$ from a noisy optical signal not designed to measure melanin is therefore a meaningful positive result.

## 3.4 Skin Tone Prediction Experiment

Using the normoxic $r_{DC}$ (specifically the $q_{10}$ percentile per encounter, the most stable estimate), a Gradient Boosted Regression (GBR) model was trained to predict the patient's Monk Skin Tone scale value (10 levels, A–J) using leave-one-out cross-validation (LOO-CV).

| **Metric** | **Value** |
|---|---|
| 4-ch Spearman $\rho$ ($r_{DC}$ vs. Monk tone) | $-0.623$ |
| Naïve mean $r_{DC}$ correlation | $-0.510$ |
| GBR Monk prediction MAE (LOO-CV, all channels) | $\approx 1.483$ steps / 10 |
| Best feature extraction method | $SaO_2 \geq 94\%$, $q_{10}$ per encounter |

Table 2: Summary of Direction 1 quantitative results.

An MAE of $\approx 1.5$ Monk steps means the model can separate broadly light from broadly dark skin, but cannot reliably distinguish adjacent tones (e.g., tone C from D). This is a fundamental limitation of using wavelengths optimized for hemoglobin rather than melanin. Additional methods explored included 10/20/30 s window comparison (20 s optimal), per-beat $r_{DC}$, respiratory modulation ratio, pulse area ratio, AC amplitude ratio, and deep learning on raw 4-channel waveforms (BiLSTM, TCN).

### 3.5 Design Decision: Use $r_{DC}$ as a Continuous Melanin Proxy

Rather than predicting a discrete Monk label — which introduces quantization error and a two-step pipeline — $r_{DC}$ will be fed directly into the $SpO_2$ correction model as a continuous melanin proxy. This allows the correction model to learn the relationship between optical skin absorption and $SpO_2$ bias end-to-end, without requiring any demographic label for race or skin tone. This finding has been incorporated into the v6 blood gas model, where $r_{DC}$ features appear alongside COHb and pH.

| **Question** | **Answer** |
|---|---|
| Is skin tone encoded in the PPG signal? | Yes — $\rho = -0.623$ on 4-channel devices |
| Why does the normoxic filter help? | Removes deoxyHb interference at 660 nm |
| Can we predict the Monk scale? | Broadly (MAE $\approx 1.5/10$ steps), not precisely |
| What is the practical outcome? | Feed $r_{DC}$ directly into the correction model as a skin tone proxy — no demographic labels needed |

Table 3: Direction 1 summary for reference.